Ps 06

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1 Problem 1 DONE

1.1 Task

Consider the function $f : [0, 1] \to \mathbb{R}$ defined by

$$f(x) = 3x + 2$$

and the equidistant partitions $\mathcal{P}_n = \{\frac{k}{n} : k = 0, ..., n\}$, for $n \in \mathbb{N}$.

(a) Determine $L(f, \mathcal{P}_n)$ and $U(f, \mathcal{P}_n)$. (b) Determine the limits $\lim_{n\to\infty} L(f, \mathcal{P}_n)$ and $\lim_{n\to\infty} U(f, \mathcal{P}_n)$. (c) Compute $\int_0^1 f(x)dx$ by the FTC.

1.2 Solution

- Part (a)
  - Setup Let $\mathcal{P}_n = \{x_0, x_1, \ldots, x_n\}$ be the equidistant partition with $x_k = \frac{k}{n}$ and $\Delta x = \frac{1}{n}$. Since $f(x) = 3x + 2$ is strictly increasing on $[0, 1]$, the infimum and supremum on each sub-interval $[x_{k-1}, x_k]$ are achieved at the endpoints:

$$\begin{aligned} m_k &= \inf_{x\in[x_{k-1},x_k]} f(x) = f(x_{k-1}) = 3\left(\frac{k-1}{n}\right) + 2 \\ M_k &= \sup_{x\in[x_{k-1},x_k]} f(x) = f(x_k) = 3\left(\frac{k}{n}\right) + 2 \end{aligned}$$

  - Lower Sum Calculation

$$\begin{aligned} L(f, \mathcal{P}_n) &= \sum_{k=1}^{n} m_k \Delta x \\ &= \sum_{k=1}^{n} \left[3\left(\frac{k-1}{n}\right) + 2\right] \frac{1}{n} \\ &= \frac{1}{n}\left[\frac{3}{n}\sum_{k=1}^{n}(k-1) + \sum_{k=1}^{n} 2\right] \\ &= \frac{1}{n}\left[\frac{3}{n}\cdot\frac{(n-1)n}{2} + 2n\right] \quad (\text{using } \sum_{i=0}^{n-1} i = \frac{(n-1)n}{2}) \\ &= \frac{3(n-1)}{2n} + 2 \\ &= \frac{3}{2} - \frac{3}{2n} + \frac{4}{2} \\ &= \frac{7}{2} - \frac{3}{2n} \end{aligned}$$

- Upper Sum Calculation

$$\begin{aligned}
U(f, \mathcal{P}_n) &= \sum_{k=1}^{n} M_k \Delta x \\
&= \sum_{k=1}^{n} \left[3\left(\frac{k}{n}\right) + 2\right] \frac{1}{n} \\
&= \frac{1}{n}\left[\frac{3}{n}\sum_{k=1}^{n} k + \sum_{k=1}^{n} 2\right] \\
&= \frac{1}{n}\left[\frac{3}{n} \cdot \frac{n(n+1)}{2} + 2n\right] \\
&= \frac{3(n+1)}{2n} + 2 \\
&= \frac{3}{2} + \frac{3}{2n} + \frac{4}{2} \\
&= \frac{7}{2} + \frac{3}{2n}
\end{aligned}$$

- Part (b)

  Determine the limits $\lim_{n\to\infty} L(f, \mathcal{P}_n)$ and $\lim_{n\to\infty} U(f, \mathcal{P}_n)$.

  Using the expressions derived in Part (a):

  - **Limit of Lower Sum**

$$\begin{aligned}
\lim_{n\to\infty} L(f, \mathcal{P}_n) &= \lim_{n\to\infty} (\frac{7}{2} - \frac{3}{2n}) \\
&= \frac{7}{2} - 0 \\
&= \frac{7}{2}
\end{aligned}$$

  - **Limit of Upper Sum**

$$\begin{aligned}
\lim_{n\to\infty} U(f, \mathcal{P}_n) &= \lim_{n\to\infty} (\frac{7}{2} + \frac{3}{2n}) \\
&= \frac{7}{2} + 0 \\
&= \frac{7}{2}
\end{aligned}$$

  Since $\lim_{n\to\infty} L(f, \mathcal{P}_n) = \lim_{n\to\infty} U(f, \mathcal{P}_n) = \frac{7}{2}$, the function is Riemann integrable on $[0, 1]$ and the value of the integral is $\frac{7}{2}$.

- Part (c)

  Compute $\int_0^1 f(x)dx$ by the FTC.

$$\int_a^b f(x)\, dx = F(b) - F(a) \quad \text{where} \quad F'(x) = f(x)$$

$$
\begin{aligned}
\int_0^1 (3x+2)\,dx &= \left[\frac{3}{2}x^2 + 2x\right]_0^1 \\
&= \left(\frac{3}{2}(1)^2 + 2(1)\right) - \left(\frac{3}{2}(0)^2 + 2(0)\right) \\
&= \frac{7}{2} - 0 \\
&= \frac{7}{2}
\end{aligned}
$$

2 Problem 2 DONE

2.1 Task

Determine the following integrals: (a) $\int_1^3 x^2 dx$ (b) $\int_0^{\log 2} e^x dx$ (c) $\int_0^{\pi} \sin x dx$ (d) $\int_1^e \frac{1}{x} dx$ In addition, approximate each integral in Julia by choosing the appropriate equidistant partition of the respective intervals and implementing $L(f, \mathcal{P}_n)$ and $U(f, \mathcal{P}_n)$.

2.2 Solution

- Part (a)

$$
\begin{aligned}
\int_1^3 x^2\,dx &= \left[\frac{1}{3}x^3\right]_1^3 \\
&= (\frac{1}{3}(3)^3) - (\frac{1}{3}(1)^3) \\
&= 9 - \frac{1}{3} \\
&= \frac{26}{3}
\end{aligned}
$$

- Part (b)

$$
\begin{aligned}
\int_0^{\log(2)} e^x\,dx &= [e^x]_0^{\log(2)} \\
&= (e^{\log(2)}) - (e^0) \\
&= 2 - 1 \\
&= 1
\end{aligned}
$$

- Part (c)

$$\begin{aligned}\int_0^{\pi} \sin(x)\,dx &= [-\cos(x)]_0^{\pi} \\ &= -\cos(\pi) + \cos(0) \\ &= 1 + 1 \\ &= 2\end{aligned}$$

- Part (d)

$$\begin{aligned}\int_1^{e} \frac{1}{x}\,dx &= [\log(x)]_1^{e} \\ &= \log(e) - \log(1) \\ &= 1 - 0 \\ &= 1\end{aligned}$$

- Code

```
function L(f, P)
    area = 0.0
    for i in 1:length(P)-1
        h = min(f(P[i]), f(P[i+1]))

        w = P[i+1] - P[i]

        area += h * w
    end
    return area
end

function U(f, P)
    area = 0.0
    for i in 1:length(P)-1
        h = max(f(P[i]), f(P[i+1]))

        w = P[i+1] - P[i]

        area += h * w
    end
    return area
end
```

```
function exercise()
    n = 1000

    # each item has: label, function, start(a), end(b), and exact answer
    data = [
        (label="(a) x^2",     f = x -> x^2,     a = 1,  b = 3,       exact = 26/3),
        (label="(b) e^x",     f = x -> exp(x),  a = 0,  b = log(2),  exact = 1   ),
        (label="(c) sin(x)",  f = x -> sin(x),  a = 0,  b = pi,      exact = 2   ),
        (label="(d) 1/x",     f = x -> 1/x,     a = 1,  b = exp(1),  exact = 1   )
    ]

    println("Approximating integrals with n = $n partitions:\n")
    println(rpad("Problem", 15), rpad("Lower Sum (L)", 18), rpad("Upper Sum (U)", 18), "E
    println("-"^65)

    for i in data
        P_n = range(i.a, i.b, length=n+1)

        val_L = L(i.f, P_n)
        val_U = U(i.f, P_n)

        print(rpad(i.label, 15))
        print(rpad(round(val_L, digits=5), 18))
        print(rpad(round(val_U, digits=5), 18))
        println(round(i.exact, digits=5))
    end

end

exercise()
```

```
L (generic function with 1 method)
U (generic function with 1 method)
exercise (generic function with 1 method)
Approximating integrals with n = 1000 partitions:

Problem        Lower Sum (L)     Upper Sum (U)     Exact
-----------------------------------------------------------------
(a) x^2        8.65867           8.67467           8.66667
(b) e^x        0.99965           1.00035           1.0
(c) sin(x)     1.99686           2.00314           2.0
(d) 1/x        0.99946           1.00054           1.0
```

3 Problem 3 DONE

3.1 Task

Determine the following integrals using integration by parts. (a) $\int_0^1 xe^x dx$ (b) $\int_0^\pi x \sin x dx$ (c) $\int_1^e \log x dx$ (d) $\int_0^1 x \log x dx$ (e) $\int_0^1 x^2 e^x dx$

In addition, implement the trapezoidal rule in Julia for equidistant partitions and compare the approximation with the exact value.

3.2 Solution

- Rules
  - Integration by Parts

$$\int u\,dv = uv - \int v\,du$$

  - Integration by Substitution

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

- (a) $\int_0^1 xe^x\,dx$

$$\begin{aligned}
u = x &\implies du = 1 \\
dv = e^x &\implies v = e^x \\
\int_0^1 xe^x\,dx &= \Big[xe^x\Big]_0^1 - \int_0^1 e^x\,dx \\
&= e - \int_0^1 e^x\,dx \\
&= e - \Big[e^x\Big]_0^1 \\
&= e - (e-1) \\
&= 1
\end{aligned}$$

- (b) $\int_0^\pi x\sin(x)\,dx$

$$
\begin{aligned}
u = x &\implies du = 1 \\
dv = \sin(x) &\implies v = -\cos(x) \\
\int_0^\pi x\sin(x)\,dx &= \Big[-x\cos(x)\Big]_0^\pi - \int_0^\pi -\cos(x)\,dx \\
&= \pi + \int_0^\pi \cos(x)\,dx \\
&= \pi + \Big[\sin(x)\Big]_0^\pi \\
&= \pi
\end{aligned}
$$

- (c) $\int_1^e \log(x)\,dx$

$$
\begin{aligned}
u = \log(x) &\implies du = \frac{1}{x} \\
dv = 1 &\implies v = x \\
\int_1^e \log(x)\,dx &= \Big[x\log(x)\Big]_1^e - \int_1^e 1\,dx \\
&= e - \int_1^e 1\,dx \\
&= e - \Big[x\Big]_1^e \\
&= e - (e-1) \\
&= 1
\end{aligned}
$$

- (d) $\int_0^1 x\log(x)\,dx$

$$
\begin{aligned}
u = \log(x) &\implies du = \frac{1}{x} \\
dv = x &\implies v = \frac{1}{2}x^2 \\
\int_0^1 x\log(x)\,dx &= \Big[\frac{1}{2}x^2\log(x)\Big]_0^1 - \int_0^1 \frac{1}{2}x\,dx \\
&= \Big[\frac{1}{2}x^2\log(x)\Big]_0^1 - \Big[\frac{1}{4}x^2\Big]_0^1 \\
&= \Big[\frac{1}{2}x^2\log(x) - \frac{1}{4}x^2\Big]_0^1 \\
&= -\frac{1}{4}
\end{aligned}
$$

- (e) $\int_0^1 x^2 e^x\,dx$

$$
\begin{aligned}
u = x^2 &\implies du = 2x \\
dv = e^x &\implies v = e^x \\
\int_0^1 x^2 e^x \, dx &= \left[x^2 e^x\right]_0^1 - \int_0^1 2xe^x \, dx \\
&= e - \int_0^1 2xe^x \, dx \\
&= e - 2\int_0^1 xe^x \, dx \\
&= e - 2 \qquad \text{(part (a))}
\end{aligned}
$$

- Code

```
using Printf

function T(f, P)
    area = 0.0
    for i in 1:length(P)-1
        h = (f(P[i]) + f(P[i+1])) / 2
        w = P[i+1] - P[i]
        area += h * w
    end
    return area
end

function exercise()
    n = 1000

    safe_xlogx(x) = (x == 0) ? 0.0 : x * log(x)

    data = [
        (label="(a) x e^x",     f = x -> x * exp(x),    a = 0.0, b = 1.0,     exact = 1.0
        (label="(b) x sin(x)",  f = x -> x * sin(x),    a = 0.0, b = pi,      exact = pi
        (label="(c) log(x)",    f = x -> log(x),        a = 1.0, b = exp(1),  exact = 1.0
        (label="(d) x log(x)",  f = safe_xlogx,         a = 0.0, b = 1.0,     exact = -0.
        (label="(e) x^2 e^x",   f = x -> x^2 * exp(x),  a = 0.0, b = 1.0,     exact = exp
    ]

    println("Approximating integrals using Trapezoidal Rule (n = $n):\n")
    println(rpad("Problem", 16), rpad("Approx (T)", 18), rpad("Exact", 18), "Diff")
    println("-"^65)

    for i in data
        P_n = range(i.a, i.b, length=n+1)
```

```
        val_T = T(i.f, P_n)
        diff = abs(val_T - i.exact)

        print(rpad(i.label, 16))
        print(rpad(round(val_T, digits=6), 18))
        print(rpad(round(i.exact, digits=6), 18))
        @printf "%.2e\n" diff
    end
end

exercise()
```

```
T (generic function with 1 method)
exercise (generic function with 1 method)
Approximating integrals using Trapezoidal Rule (n = 1000):

Problem         Approx (T)        Exact             Diff
-----------------------------------------------------------------
(a) x e^x       1.0               1.0               3.70e-07
(b) x sin(x)    3.14159           3.141593          2.58e-06
(c) log(x)      1.0               1.0               1.56e-07
(d) x log(x)    -0.249999         -0.25             8.24e-07
(e) x^2 e^x     0.718283          0.718282          6.80e-07
```

4 Problem 4 DONE

4.1 Task

Determine the following integrals using an appropriate substitution: (a) $\int_0^1 2xe^{x^2}dx$ (b) $\int_0^{\pi/2} \sin(x)\cos(x)dx$ (c) $\int_1^2 \frac{2x}{1+x^2}dx$ (d) $\int_0^{\log 2} e^{2x}dx$ (e) $\int_0^1 \frac{1}{\sqrt{1-x^2}}dx$

In addition, implement the mid-point rule in Julia for equidistant partitions and compare the approximation with the exact value.

4.2 Solution

- Rules
  - Integration by Parts

$$\int u\,dv = uv - \int v\,du$$

  - Integration by Substitution

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

- (a) $\int_0^1 2xe^{x^2}\,dx$

$$\begin{aligned} f(x) = e^x &\implies f'(x) = e^x \\ g(x) = x^2 &\implies g'(x) = 2x \\ \int_0^1 2xe^{x^2}\,dx &= \left[e^{x^2}\right]_0^1 \\ &= e^1 - e^0 \\ &= e - 1 \end{aligned}$$

- (b) $\int_0^{\pi/2} \sin(x)\cos(x)\,dx$

$$\begin{aligned} f(x) = \frac{1}{2}x^2 &\implies f'(x) = x \\ g(x) = \sin(x) &\implies g'(x) = \cos(x) \\ \int_0^{\pi/2} \sin(x)\cos(x)\,dx &= \left[\frac{1}{2}(\sin(x))^2\right]_0^{\pi/2} \\ &= \left[\frac{1}{2}(\sin(x))^2\right]_0^{\pi/2} \\ &= \frac{1}{2} \end{aligned}$$

- (c) $\int_1^2 \frac{2x}{1+x^2}\,dx$

$$\begin{aligned} f(x) = \log(x) &\implies f'(x) = \frac{1}{x} \\ g(x) = 1 + x^2 &\implies g'(x) = 2x \\ \int_1^2 \frac{2x}{1+x^2}\,dx &= \left[\log(1+x^2)\right]_1^2 \\ &= \log(5) - log(2) \end{aligned}$$

- (d) $\int_0^{\log 2} e^{2x}\,dx$

$$
\begin{aligned}
f(x) = e^x &\implies f'(x) = e^x \\
g(x) = 2x &\implies g'(x) = 2 \\
\int_0^{\log 2} e^{2x}\,dx &= \frac{1}{2}\Big[e^{2x}\Big]_0^{\log 2} \\
&= \frac{1}{2}(e^{2\log(2)} - e^0) \\
&= \frac{1}{2}(4-1) \\
&= \frac{3}{2}
\end{aligned}
$$

- (e) $\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx$

$$
\begin{aligned}
f(x) = \sin(x) &\implies f'(x) = \frac{1}{\sqrt{1-x^2}} \\
\int_0^1 \frac{1}{\sqrt{1-x^2}}\,dx &= \Big[\sin^{-1}(x)\Big]_0^1 \\
&= \frac{\pi}{2}
\end{aligned}
$$

  - Notes

$$
\begin{aligned}
f(x) &= \cos^{-1}(x) \\
\cos(f(x)) &= x \\
\frac{d}{dx}\cos(f(x)) &= \frac{d}{dx}x \\
-\sin(f(x))\,\frac{dx}{dy} &= 1 \\
-\sqrt{1-\cos(f(x))^2}\,\frac{dx}{dy} &= 1 \qquad (-\sin(x) = -\sqrt{1-\cos(x)^2}) \\
-\sqrt{1-x^2}\,\frac{dx}{dy} &= 1 \\
\frac{dx}{dy} &= -\frac{1}{\sqrt{1-x^2}}
\end{aligned}
$$

- Code

```
using Printf
```

```
function M(f, P)
    area = 0.0
    for i in 1:length(P)-1
        mid = (P[i+1] + P[i]) / 2
        h = f(mid)
        w = P[i+1] - P[i]
        area += h * w
    end
    return area
end

function exercise()
    n = 1000

    data = [
        (label="(a) 2x e^(x^2)",   f = x -> 2x * exp(x^2),     a = 0.0, b = 1.0,    exact
        (label="(b) sin(x)cos(x)", f = x -> sin(x)*cos(x),     a = 0.0, b = pi/2,   exact
        (label="(c) 2x/(1+x^2)",   f = x -> (2x)/(1+x^2),      a = 1.0, b = 2.0,    exact
        (label="(d) e^(2x)",       f = x -> exp(2x),           a = 0.0, b = log(2), exact
        (label="(e) 1/sqrt(1-x^2)",f = x -> 1/sqrt(1-x^2),     a = 0.0, b = 1.0,    exact
    ]

    println("Approximating integrals using Midpoint Rule (n = $n):\n")
    println(rpad("Problem", 18), rpad("Approx (M)", 18), rpad("Exact", 18), "Diff")
    println("-"^65)

    for i in data
        P_n = range(i.a, i.b, length=n+1)

        val_M = M(i.f, P_n)
        diff = abs(val_M - i.exact)

        print(rpad(i.label, 18))
        print(rpad(round(val_M, digits=6), 18))
        print(rpad(round(i.exact, digits=6), 18))
        @printf "%.2e\n" diff
    end
end

exercise()


M (generic function with 1 method)
exercise (generic function with 1 method)
Approximating integrals using Midpoint Rule (n = 1000):

Problem           Approx (M)        Exact             Diff
-----------------------------------------------------------------
```

(a) 2x e^(x^2)	1.718281	1.718282	5.96e-07
(b) sin(x)cos(x)	0.5	0.5	2.06e-07
(c) 2x/(1+x^2)	0.916291	0.916291	1.00e-08
(d) e^(2x)	1.5	1.5	1.20e-07
(e) 1/sqrt(1-x^2)	1.557271	1.570796	1.35e-02